

No. of Functional Features Included in the Solution

Date:	30 June 2026
Team ID:	XXXXXX
Project Name:	Voice-Based concept Understanding Analyser
Maximum Marks:	5 Marks

Functional Features

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks
1	User Registration & login	Secure user authentication and login to access the platform.	Authentication Module	Done	High
2	Audio Upload & Recording	Allows users to upload or record voice explanations.	Audio Input Module	Done	High
3	Speech-to-Text Transcription	Converts spoken audio into text using OpenAI Whisper.	Whisper Engine	Done	High
4	Semantic Similarity Analysis	Compares transcript with the reference concept using Sentence-BERT.	Semantic Analysis Module	Done	High
5	Audio Feature Extraction	Extracts pause ratio, RMS energy, speech rate, and filler words using	Audio Analysis Module	Done	High

		Librosa. letters using Google Gemini AI			
6	AI Evaluation & Scoring	Generates concept understanding score and qualitative feedback.	Evaluation Engine	Done	High
7	Dashboard & Report Generation	Displays transcript, scores, graphs, and generates downloadable PDF reports.	Dashboard & Report Module	Done	High
8	Data Storage & History	Stores user information, transcripts, evaluation results, and previous reports.	SQLite Database	Done	Medium

Feature Category Breakdown:

S.NO	Category	Features in Category	Example Features
1	User Interface (UI)	2	Login Interface, Audio Upload & Dashboard
2	Backend/Logic	4	Whisper Transcription, Sentence-BERT Analysis, Librosa Audio Processing, Evaluation Engine
3	Database/Storage	1	SQLite Database, User Records, Reports, Transcripts
4	AI /External libraries	3	OpenAI Whisper, Sentence-BERT, Librosa
5	Security/Authentication	1	Secure User Login & Authentication